Sumit Dhakad

AI & Data Science Engineer II | Generative & Agentic AI Specialist
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Portfolio | LinkedIn | GitHub | HackerRank

SUMMARY

AI & Data Science Engineer II at Deloitte with 4+ years of experience architecting production-grade Agentic AI systems, Multi-Agent orchestration (LangGraph/Google-ADK), and enterprise RAG pipelines. Proven track record in scaling cloud data architectures (100GB+ on BigQuery/GCP) and deploying low-latency, air-gapped and cloud LLM applications

TECHNICAL SKILLS

Generative & Agentic AI: LangGraph, LangChain, Google-ADK, RAG Architectures, Multi-Agent Systems, Hugging Face, Ollama, LLM Fine-Tuning, Prompt Optimization, vLLM, MCP
Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-Learn, OpenCV, Computer Vision, NLP, Reinforcement Learning, MLflow
Languages & Backend: Python, SQL, FastAPI, Pydantic, REST APIs, WebSockets, JWT/RBAC Auth, Web Scraping
Databases & Vector Stores: PostgreSQL (pgvector), Redis, MinIO (S3), BigQuery, Qdrant, ChromaDB, FAISS, MongoDB
Cloud, MLOps & Tools: GCP, AWS, Docker, Kubernetes, Nginx, CI/CD, GitHub Actions, Google Cloud Storage (GCS), Linux

EXPERIENCE

Deloitte | Pune, India

AI and Data Science Engineer - II | January 2026 – Present

- Architected an AI-powered **business intelligence and partner discovery platform** using **Python, LangGraph, Google-ADK, and FastAPI**, automating organization discovery, qualification, and research to reduce manual effort by **~60%**.
- Engineered two scalable data ingestion and enrichment pipelines processing **100+ GB** of enterprise and public market data into **BigQuery**, continuously enriching organization profiles from web, social, and internal data sources.
- Designed a **Hybrid RAG architecture** combining **BigQuery and Vector Databases (Qdrant)** with **Semantic Caching and Redis TTL general caching**, reducing redundant LLM API calls by **45%**, eliminating repetitive data collection, and improving retrieval accuracy.
- Built a stateful **Multi-Agent AI Assistant** using **LangGraph, MCP, and WebSockets** (served behind an **Nginx** reverse proxy) to reason across structured, unstructured, and live web data, streaming contextual insights in real time.
- Automated service deployment using **GitHub Actions CI/CD**, integrating **Pytest** unit testing and **Docker** containerization for reliable releases on **GCP**.

Verge Impex | Mumbai, India

Machine Learning Engineer - II | November 2024 – January 2026

- Engineered the backend and AI microservices using **FastAPI** for a centralized, on-premise **Healthcare & Clinical Management Platform**, implementing **JWT authentication with refresh tokens, session management, and Role-Based Access Control (RBAC)** across Patient, Doctor, and Admin endpoints.
- Built dual **Context-Aware AI Assistants** (for Patients and Doctors) hosted locally via **vLLM** and streamed over **WebSockets**, leveraging longitudinal medical histories, past symptoms, and monthly checkups to assist doctors with prescriptions and auto-generate personalized diet, exercise, and precaution plans.
- Developed a **Vision-LLM document extraction pipeline** to parse patient-uploaded X-rays, lab reports, and prescriptions (stored in **MinIO**) into structured database schemas for continuous health tracking.
- Designed a hybrid data and infrastructure layer using **PostgreSQL**, **pgvector** for semantic medical search, and **Redis** caching, containerized with **Docker**, **Nginx** reverse proxy, and **CI/CD** for seamless local server deployment.

WESEE (Ministry of Defence / Indian Navy) | New Delhi, India

AI Engineer | July 2024 – October 2024

- Engineered an automated **Open-Source Intelligence (OSINT)** ingestion and NLP extraction pipeline processing **10,000+ daily unstructured defense and news feeds**, converting raw text into structured entity schemas with **89% extraction F1-score**.
- Deployed an autonomous **RAG and Agentic query engine** using **ChromaDB and open-weights LLMs** to analyze aggregated geopolitical intelligence, reducing analyst threat-assessment turnaround time from hours to **under 30 seconds**.

AIEnsured Pvt Ltd | Noida, India

Machine Learning Engineer | June 2022 – July 2024

- Built end-to-end scalable **MLOps pipelines** (data preprocessing, distributed training, **MLflow** tracking, and deployment), improving model inference throughput by **15%** and cutting training costs by **25%**.
- Developed a synthetic tabular and structural data generation framework achieving **85% statistical fidelity**, overcoming

client data-scarcity bottlenecks and accelerating model prototyping by **3x**.

- Designed production **Retrieval-Augmented Generation (RAG)** pipelines for enterprise document Q&A using **FAISS, LangChain, and Hugging Face** transformers, alongside interactive executive dashboards to drive data-backed decisions.

PROJECTS

Sentiment Analysis of Articles

- Preprocessed large text datasets and trained an **LSTM** baseline (**78% accuracy**), before fine-tuning a pre-trained **BERT (Hugging Face Transformer)** model to achieve **85% classification accuracy**.
- Developed a **FastAPI** backend for real-time sentiment inference and automated article categorization, containerized with **Docker** for scalable deployment.

Road Scenario Object Detection

- Developed a real-time object detection system using **YOLO and OpenCV** to identify and track vehicles and road objects across camera, image, and video feeds, achieving **85% precision and 87% recall**.
- Built an interactive **Streamlit** web application for live video and image inference to assist with road safety monitoring.

EDUCATION

B.Tech in Computer Science (Specialization in Data Science) | Aug 2018 – May 2022
ITM University, Gwalior, India

CERTIFICATIONS

- **Claude Certified Developer - Foundations** | Anthropic (2025)
  Credential ID: 2cc50f6b-0c6c-4a84-9546-ea31f70bc199
- **AWS Academy Graduate - Cloud Foundations** | Amazon Web Services (2021)
  Credential ID: abf10f79-5abe-4dd6-b558-84f5e83290b7
- **Machine Learning & Deep Learning Specializations** | Nelumbus Educare (2020 – 2021)